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Določiti enačbo elipsoida s temeni:

$$T_{1,2} = \pm \frac{\sqrt{3}}{9} (2, 2, -1) \rightarrow v_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$$

$$T_{3,4} = \pm \frac{\sqrt{6}}{18} (-1, 2, 2) \rightarrow v_2 = \frac{1}{3} \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$$

$$T_{5,6} = \pm \frac{1}{9} (2, -1, 2) \rightarrow v_3 = \frac{1}{3} \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

$$B = \frac{1}{3} \begin{bmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{bmatrix}$$

$$[T_i]_B = ?$$

$$T_{1,2} = \pm \frac{\sqrt{3}}{3} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$T_{3,4} = \pm \frac{\sqrt{6}}{6} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$T_{5,6} = \pm \frac{1}{3} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda_1 x'^2 + \lambda_2 y'^2 + \lambda_3 z'^2 = 1$$

$$\lambda_1 \left(\frac{\sqrt{3}}{3} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right)^2 = 1 \rightarrow \lambda_1 = 3$$

$$\lambda_2 = 6$$

$$\lambda_3 = 9$$

$$\begin{aligned} A &= B \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} B^T = \frac{1}{3} \begin{bmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 2 & -1 \\ -1 & 2 & 2 \\ 2 & -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -2 & 6 \\ 2 & 4 & -3 \\ -1 & 4 & 6 \end{bmatrix} \begin{bmatrix} 2 & 2 & -1 \\ -1 & 2 & 2 \\ 2 & -1 & 2 \end{bmatrix} = \\ &= \frac{1}{3} \begin{bmatrix} 18 & -6 & 6 \\ -6 & 15 & 0 \\ 6 & 0 & 21 \end{bmatrix} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 5 & 0 \\ 2 & 0 & 7 \end{bmatrix} \end{aligned}$$

$$\underline{6x^2 + 5y^2 + 7z^2 - 4xy + 4xz = 1}$$

⑤ Na $\mathbb{R}^2$ definirame bil. formo

$$\langle x, y \rangle = x_1 y_1 - x_2 y_2$$

Poišči vse matrice $A \in \mathbb{R}^{2 \times 2}$, za katere velja $\langle Ax, Ay \rangle = \langle x, y \rangle$ za $\forall x, y \in \mathbb{R}^2$

$$I_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\langle Ax, Ay \rangle = \langle I_{1,1} Ax, Ay \rangle_{std} = \langle A^T I_{1,1} A x, y \rangle_{std}$$

$$\langle x, y \rangle = \langle I_{1,1} x, y \rangle_{std}$$

$$O(1,1) = \{ A \in \mathbb{R}^{2 \times 2} ; A^T I_{1,1} A = I_{1,1} \}$$

(neki fun facti)

$$A = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$$

$\vdots$

$$x_{11}^2 - x_{21}^2 = 1$$

$$x_{11} x_{12} - x_{21} x_{22} = 0$$

$$x_{22}^2 - x_{12}^2 = 1$$

$$x_{12} = \sinh \alpha$$

$$x_{21} = \sinh \alpha'$$

$\Downarrow$

$$x_{11} = \pm \cosh \alpha$$

$$x_{22} = \pm \cosh \alpha'$$

$\Downarrow$

$$\alpha' = \pm \alpha$$

$$\rightarrow H_\alpha = \begin{bmatrix} \cosh \alpha & -\sinh \alpha \\ -\sinh \alpha & \cosh \alpha \end{bmatrix}$$

hiperbol. rot.

Funkc. več spr.

stac. točke: $\nabla f = 0$

$H > 0 \Rightarrow$ lok. minimum

$H < 0 \Rightarrow$ lok. maximum

H obrnjiva, nedefinitna $\Rightarrow$ ni ekstremna

sicer pogl. višje odrede

Sylvester:

$H > 0$: vse gl. poddet. > 0

$H < 0$: sode gl. poddet. > 0
lihe gl. poddet. < 0

① Poišči in klas. lok. ekstr. funke.

$$f(x, y, z) = 2x^2 - 2xy + 2y^2 + 4xz + 2yz + 5z^2 - 2x + 2y + 1$$